

JOINT DISTRIBUTION-INFORMED SHAPLEY VALUES FOR SPARSE COUNTERFACTUAL EXPLANATIONS

Anonymous authors
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ABSTRACT

Counterfactual explanations (CE) aim to reveal how small input changes flip a model’s prediction, yet many methods modify more features than necessary, reducing clarity and actionability. We introduce *COLA*, a model- and generator-agnostic post-hoc framework that refines any given CE by computing a coupling via optimal transport (OT) between factual and counterfactual sets and using it to drive a Shapley-based attribution (*p*-SHAP) that selects a minimal set of edits while preserving the target effect. Theoretically, OT minimizes an upper bound on the W_1 divergence between factual and counterfactual outcomes, and, under mild conditions, refined counterfactuals are guaranteed not to move farther from the factuals than the originals. Empirically, we benchmark COLA against five competing attribution schemes across a broad grid of four datasets, twelve models, and five CE generators. The gains are large and consistent: to reproduce the same target effect, COLA edits only 26–45% of the features changed by the original CE, and its *p*-SHAP variant is the single method that attains the full (100%) target effect on *every* dataset while every baseline fails to reach it. Against the competing Shapley couplings, *p*-SHAP sharply cuts the number of edits (e.g., on Hotel Bookings at the 80% effect level it needs 2.50 features versus 7.01–10.63 for the CF baselines), and it is the best method in every evaluated scenario. On a small-scale benchmark where the global optimum is computable, COLA lands within a narrow margin of the exact MILP solution.

The experiment code and software will be released upon acceptance.

1 BACKGROUND

Explainable Artificial Intelligence (XAI) is essential for making artificial intelligence systems transparent and trustworthy (Arrieta et al., 2020; Das & Rad, 2020). Within this area, feature attributions (FA) methods, such as Shapley values (Sundararajan & Najmi, 2020; Lundberg & Lee, 2017), quantify how much each input feature contributes to a machine learning (ML) model’s output, and thereby simplify complex models by ranking the most influential features. In a healthcare model, for instance, Shapley values can single out key factors such as age and medical history, helping clinicians understand a decision (Ter-Minassian et al., 2023; Nohara et al., 2022). A complementary technique, counterfactual explanations (CE) (Wachter et al., 2017; Guidotti, 2022), shows how small changes in input features can flip an outcome. Although hundreds of CE algorithms have been proposed (Guidotti, 2022; Verma et al., 2020), no single algorithm serves every use case, because each is tailored to a specific scenario, goal, and task. Some algorithms find a single counterfactual for each factual instance; others treat several instances as a group and return one or more counterfactuals per factual observation. Some target the entire dataset, seeking global CE that indicate the direction in which factual instances should move to obtain the desired output (Rawal & Lakkaraju, 2020; Ley et al., 2022; 2023; Carrizosa et al., 2024). Others treat the group of factual instances as a distribution and search for a counterfactual distribution that preserves the factual shape at comparable cost (You et al., 2025). Some assume differentiable models, while others target tree-based or ensemble models. This diversity is precisely why we design and evaluate COLA as a generator-agnostic refinement layer: it must deliver consistent quantitative gains regardless of which of these regimes produced the CE.

Problem Description and Challenges Although both FA and CE are vital for making AI models more interpretable and accountable, relying on only one of them leaves a measurable gap. FA alone offers no tangible action steps, while CE alone tends to change far more features than needed—a gap that our experiments later quantify precisely, showing that most of the edits proposed by standard CE generators are removable without loss of counterfactual effect. Motivated by this gap, we address a general and comprehensive problem in this paper, which builds on the extensive foundations established in the literature (see Appendix A). To be specific, we seek to answer the following question:

Given a (group of) factual instance(s), how can we devise an action plan that requires the least feature modifications to achieve a desired counterfactual outcome?

$\mathbf{x}$			$\mathbf{z}'$			$\mathbf{z}''$			$\mathbf{y}^*$
Ⓞ	Ⓜ	Ⓡ	Ⓞ	Ⓜ	Ⓡ	Ⓞ	Ⓜ	Ⓡ	Ⓡ
200	5	No	250	8	Yes	200	5	No	Yes
150	3	No	150	3	No	150	7	Yes	Yes
100	2	No	350	9	Yes	100	2	No	Yes
150	6	No	150	6	No	350	6	Yes	Yes

Figure 1: [Example: User engagement on an e-commerce platform] A platform aims to increase user registrations, having collected data on user interactions such as money spent (Ⓞ), number of clicks (Ⓜ), and registration status (Ⓡ). In the original data ($\mathbf{x}$), no users are registered. Action plans $\mathbf{z}'$ and $\mathbf{z}''$ both adjust user characteristics to reach the same desired outcome ($\mathbf{y}^*$) of full registration and both attain an identical half counterfactual effect—so the plans are indistinguishable on effect, and the only axis on which they differ is edit count: $\mathbf{z}''$ achieves the identical effect with strictly fewer modifications than $\mathbf{z}'$. This margin in edit count, at matched effect, is exactly the quantity COLA is built to minimize.

Three major challenges remain in addressing this problem. First, it is unrealistic to expect a single CE algorithm to meet all needs universally, as the problem is often task-specific; a useful method must therefore be benchmarked across many generators rather than tuned to one. Second, the approach should not rely on strong assumptions about the model (for example, differentiability or special structure), so that it applies across a wide range of models—which is why our evaluation spans twelve classifiers. Third, generic FA such as feature importance can be misleading, because the FA scores need not cohere with the changes that actually produce the counterfactual effect. In other words, it is ineffective to perform FA independently of CE to select the most important features to change. We quantify this decoupling directly: in Result II of Section 6 we show that the CE-agnostic baselines (RB- p_{Uni} and RB- p_{OT}) are the two worst performers and frequently fail to reach the target effect at all, confirming that features deemed important in general do not align with the specific pathways that achieve the desired counterfactual outcome.

Main Contributions We introduce COUNTERFACTUALS WITH LIMITED ACTIONS (COLA), a general post-hoc framework that refines CE across models and CE generators by using an optimal transport (OT)–induced coupling between factual and counterfactual sets to guide Shapley attributions. This yields p -SHAP, which unifies other commonly used Shapley methods under appropriate couplings and offers a modular interface for attribution and edit selection. We provide theoretical guarantees: OT minimizes an upper bound on the W_1 divergence, and under mild conditions refined counterfactuals are provably no farther from factuals than the originals. Empirically, the evidence is broad and one-sided. Over a grid of four datasets, twelve models, and five CE methods, COLA with p -SHAP reproduces the same target effect while editing only 26–45% of the features the original CE changed—a $2\times$ to nearly $4\times$ reduction. In head-to-head comparisons against five alternative attribution schemes, p -SHAP (CF- p_{OT}) is the best method in every scenario reported in Table 3, and it is the *only* method that reaches the full 100% counterfactual effect on all four datasets. On a small-scale benchmark where the exact optimum is computable via mixed integer linear programming (MILP), COLA tracks that optimum closely.

To our best knowledge, this is the first method to systematically address the problem in Figure 1 *without committing to a particular CE algorithm or model*, and—as the comparisons in Section 6

make quantitative—the first to do so with uniformly favorable edit-count margins across every dataset, model, and generator we test.

2 PROBLEM FORMULATION

We formally formulate the problem described in Figure 1. Denote $f : \mathbb{R}^d \rightarrow \mathbb{R}$ as a black-box ML model. Denote by $\mathbf{x}$ any observed (factual) data with n rows and d columns ($\mathbf{x} \in \mathbb{R}^{n \times d}$, $n \geq 1$, and $d \geq 1$). Let $\mathbf{y}^*$ be the target model output ($\mathbf{y}^* \in \mathbb{R}^m$, $m \geq 1$). The optimization is to look for a (group of) counterfactual data instance(s) $\mathbf{z}$ ($\mathbf{z} \in \mathbb{R}^{n \times d}$) subject to a maximum number of allowed feature changes, C , to achieve model output(s) as close as possible to $\mathbf{y}^*$. Let D denote a divergence function that measures the dissimilarity between two entities. The problem is below.

$$\min_{\mathbf{c}, \mathbf{z}} D(f(\mathbf{z}), \mathbf{y}^*), \quad (1a)$$

$$\text{s.t. } D(\mathbf{z}, \mathbf{x}) \leq \epsilon, \quad (1b)$$

$$\sum_{i=1}^n \sum_{k=1}^d c_{ik} \leq C, \quad (1c)$$

$$z_{ik} \leq x_{ik}(1 - c_{ik}) + M c_{ik} \quad i = 1, \dots, n, \quad k = 1, \dots, d, \quad (1d)$$

$$z_{ik} \geq x_{ik}(1 - c_{ik}) - M c_{ik} \quad i = 1, \dots, n, \quad k = 1, \dots, d. \quad (1e)$$

The objective equation (1a) together with the constraint equation (1b) formulate the typical CE optimization: $\mathbf{z}$ should drive $f(\mathbf{z})$ close to $\mathbf{y}^*$ while remaining close to $\mathbf{x}$, so that $\mathbf{z}$ serves as a contrastive reference explaining why $f(\mathbf{x})$ does not reach $\mathbf{y}^*$. We deliberately keep the divergence D general—it may be Euclidean distance, OT, maximum mean discrepancy (MMD), or a mean/median difference of model outcomes—so that the same edit-count metric can be compared consistently across all four of these divergences in our experiments. Constraints equation (1c)–equation (1e) then turn this into an *action-constrained* CE problem: alongside $\mathbf{z}$ we optimize an indicator $\mathbf{c}$, where z_{ik} may change iff $c_{ik} = 1$, and at most C changes are permitted by equation (1c). Reading equation (1d)–equation (1e): if $c_{ik} = 0$ then $x_{ik} = z_{ik}$ and position (i, k) is untouched; if $c_{ik} = 1$, the sufficiently large constant M frees z_{ik} to move. The count $\sum_{ik} c_{ik}$ is thus the quantity—number of edits at a fixed target effect—on which every method in Section 6 is scored.

To solve equation (1), we resort to FA to identify the most influential features and thereby set the modification indicator $\mathbf{c}$. The next section reviews commonly used Shapley value methods for FA, which—together with our proposed p -SHAP—are integrated into the COLA framework to produce the refined counterfactual $\mathbf{z}$. This detour through FA is not merely convenient: the problem is computationally hard even in the simplest case $d = 1$ with linear models (Appendix B), so the practical yardstick for any tractable method is how closely it approaches the exact optimum—a comparison we report explicitly in Result III.

3 PRELIMINARIES ON SHAPLEY VALUE

This section introduces commonly used Shapley value methods for FA, which, together with our later proposed one, are integrated into our algorithmic framework COLA and serve as the baselines against which p -SHAP is measured in Section 6. We first recall the game-theoretic Shapley value, then discuss the three common FA instantiations—B-SHAP, RB-SHAP, and CF-SHAP—that later appear as the comparison points in Table 2. Readers may consult (Sundararajan & Najmi, 2020) for a comprehensive overview.

In cooperative game theory, a coalitional game is characterized by a finite set of players (in our context, features), denoted by $\mathcal{F} = \{1, 2, \dots, d\}$, and a characteristic function v . This function $v : 2^{\mathcal{F}} \rightarrow \mathbb{R}$ maps each subset $S \subseteq \mathcal{F}$ to a real number $v(S)$, representing the total payoff that can be achieved by the members of S through cooperation, with the condition $v(\emptyset) = 0$. The Shapley value, as defined based on v later, is a fundamental concept in this framework, providing an equitable method to distribute the overall payoff, $v(\mathcal{F})$, of the grand coalition among the individual players.

Prior to defining the Shapley value, denote $\Delta(k, S)$ the incremental value produced by a player (feature) k to a coalition S , aligning with the concept of marginal contribution. This is defined as the additional payoff realized by the coalition due to the inclusion of the player k , i.e.,

$$\Delta(k, S) = v(S \cup k) - v(S). \quad (2)$$

The Shapley value ϕ_i for a player i in a cooperative game is defined in equation (3) below.

$$\phi_k = \frac{1}{d} \sum_{S \subset \mathcal{F} \setminus \{k\}} \binom{d-1}{|S|}^{-1} \Delta(k, S). \quad (3)$$

The value ϕ_k is computed by averaging the normalized marginal contributions $\Delta(k, S)$ of feature k across all subsets S of the feature set excluding k . The normalization factor is the inverse of the binomial coefficient $\binom{d-1}{|S|}$, where d is the total number of features. By summing these values for all subsets and dividing by d , the equation provides a fair and axiomatic distribution of payoffs among players, reflecting each features's contribution among the whole collection of features.

For an arbitrary data point $\mathbf{x}_i$, we use $\mathbf{x}_{i,S}$ to represent its part containing only the features in S , and $\mathbf{x}_{i,\mathcal{F} \setminus S}$ the other remaining part. The same rule applies for other notations as well. Below, we introduce commonly used Shapley value methods.

Baseline Shapley (B-SHAP). This approach relies on a specific baseline data point $\mathbf{r}_j$ as the reference for any $\mathbf{x}_i$. The set function is defined to be

$$v_B^{(i)}(S) = f(\mathbf{x}_{i,S}; \mathbf{r}_{j,\mathcal{F} \setminus S}) - f(\mathbf{r}_j), \quad (4)$$

where a feature's absence is modeled using its value in the reference baseline data point $\mathbf{r}$ (Lundberg & Lee, 2017; Sun & Sundararajan, 2011; Merrick & Taly, 2020). This method assumes an exact alignment between $\mathbf{x}$ and $\mathbf{r}$.

Random Baseline Shapley (RB-SHAP). This approach is a variant of BShap and is implicitly used in (Lundberg & Lee, 2017) as well as the famous "SHAP" library. The equation is as follows.

$$v_{RB}^{(i)}(S) = \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}} [f(\mathbf{x}_{i,S}; \mathbf{x}'_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{x}' \sim \mathcal{D}} [f(\mathbf{x}')]. \quad (5)$$

where $\mathcal{D}$ is the background distribution that is commonly selected to be the training dataset (Lundberg & Lee, 2017; Merrick & Taly, 2020).

Counterfactual Shapley (CF-SHAP). This is an extension from RB-SHAP, by taking the distribution $\mathcal{D}$ as the counterfactual distribution defined on every single factual instance $\mathbf{x}_i$ ($i = 1, 2, \dots, n$).

$$v_{CF}^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim \mathcal{D}(\mathbf{x}_i)} [f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{r} \sim \mathcal{D}(\mathbf{x}_i)} [f(\mathbf{r})]. \quad (6)$$

This method, adopted in (Albini et al., 2022; Kommiya Mothilal et al., 2021), has shown advantages for contrastive FA by conditioning on counterfactual information; it assumes a probabilistic alignment between $\mathbf{x}$ and $\mathbf{r}$. As our experiments make quantitative, however, merely conditioning on counterfactuals is not enough: the CF-SHAP variants that use uniform or random couplings (CF- p_{Uni} , CF- p_{Rnd}) are decisively outperformed by the OT-coupled p -SHAP.

4 THE PROPOSED JOINT-PROBABILITY-INFORMED SHAPLEY (p -SHAP) AND ITS THEORETICAL ASPECTS

(Proposed) p -SHAP We generalize equation (4) to equation (6) by integrating an algorithm A_{Prob} that returns their joint probability.¹ Our p -SHAP is defined as follows,

$$v^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r}|\mathbf{x}_i)} [f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r})} [f(\mathbf{r})], \quad (7a)$$

$$\text{s.t. } p = A_{\text{Prob}}(\mathbf{x}, \mathbf{r}). \quad (7b)$$

Crucially, unlike prior methods that rely on random baselines, p -SHAP solves the alignment problem by identifying the optimal coupling via OT. This reformulates feature attribution not merely as a prediction decomposition, but as a transport problem that minimizes the cost of explanation.

¹First, p -SHAP degrades to B-SHAP in equation (4) when A_{Prob} defines a joint distribution between $\mathbf{x}$ and $\mathbf{r}$ that indicates an $i \leftrightarrow j$ alignment of for any $\mathbf{x}_i, \mathbf{r}_j$. Second, p -SHAP degrades to RB-SHAP in equation (5) when A_{Prob} is defined to be independent of CE but associates with an arbitrary distribution $\mathcal{D}$. Third, p -SHAP degrades to CF-SHAP in equation (6) when A_{Prob} is built upon a known distribution of CE.

Remark that A_{Prob} is independent of the CE algorithm and depends only on the generated counterfactual $\mathbf{r}$. By depending only on these fixed components, p -SHAP keeps FA consistent across the variability of different CE generation processes—the key point of departure from CF-SHAP—and this consistency is exactly what our cross-generator, cross-model benchmark is designed to test. Contrary to common expectations, we show that the OT coupling is *more* effective than a counterfactual distribution defined by a CE generation mechanism as in [Albini et al. \(2022\)](#): in Result II of Section 6, CF- p_{OT} consistently beats every CE-mechanism-based competitor, and in Result III it even matches or surpasses the exact-alignment variant CF- p_{Ect} .

A central instance considered in this paper is the OT problem (equivalently, the squared 2-Wasserstein divergence) defined below, whose transportation plan $\mathbf{p}_{\text{OT}}$ we use as the joint distribution of $\mathbf{x}$ and $\mathbf{r}$ in p -SHAP. This is the specific coupling that, in every scenario of Section 6, yields the lowest edit counts among the six methods compared.

$$\mathbf{p}_{\text{OT}} = \arg \min_{\mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 + \varepsilon \sum_{i=1}^n \sum_{j=1}^m p_{ij} \log \left(\frac{p_{ij}}{\mu_i \nu_j} \right). \quad (8)$$

Note that $\boldsymbol{\mu}$ and $\boldsymbol{\nu}$ represent the marginal distributions of $\mathbf{x}$ and $\mathbf{r}$ respectively, and $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$ the set of joint distributions (i.e. all possible transport plans). The term $\varepsilon \sum_{i=1}^n \sum_{j=1}^m p_{ij} \log(p_{ij}/(\mu_i \nu_j))$ is the entropic regularization with $\varepsilon \geq 0$ being the coefficient. Such regularization ($\varepsilon > 0$) helps accelerate the computation of OT.

Theoretical Aspects of p -SHAP OT minimizes the total feature-modification cost of moving $\mathbf{x}$ towards $\mathbf{r}$ under the alignment it induces between factual $\mathbf{x}$ and counterfactual $\mathbf{r}$, which is exactly our objective of achieving the counterfactual outcome at minimal cost. We make this connection precise under a Lipschitz-continuity assumption on f . Theorem 4.1 below (proof in Appendix C) shows that the plan $\mathbf{p}_{\text{OT}}$ used in p -SHAP minimizes an upper bound on the divergence between $f(\mathbf{x})$ and $\mathbf{y}^*$: the 1-Wasserstein distance between $f(\mathbf{x})$ and $\mathbf{y}^*$ is at most the Lipschitz constant L times the square root of the minimized expected transport cost $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$, where each p_{ij} ($j = 1, \dots, m$) quantifies how much $\mathbf{x}_i$ should move towards $\mathbf{r}_j$. The practical reading is comparative: among all admissible plans $\mathbf{p} \in \Pi$, $\mathbf{p}_{\text{OT}}$ attains the *smallest* expected cost $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$ and hence the *tightest* upper bound on the counterfactual-effect violation $W_1(f(\mathbf{x}), \mathbf{y}^*)$. In other words, no alternative coupling can certify a smaller worst-case effect gap, which foreshadows the empirical dominance of CF- p_{OT} over the uniform, random, and RB couplings reported later.

Theorem 4.1 (p -SHAP Towards Counterfactual Effect). *Consider the 1-Wasserstein divergence W_1 , i.e. $W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*|$. Suppose the counterfactual outcome $\mathbf{y}^*$ is fully achieved by $\mathbf{r}$, i.e. $\mathbf{y}_j^* = f(\mathbf{r}_j)$ ($j = 1, 2, \dots, m$). Assume that the model $f: \mathbb{R}^d \rightarrow \mathbb{R}$ is Lipschitz continuous with Lipschitz constant L . The expected absolute difference in model outputs between the factual and counterfactual instances, weighted by $\mathbf{p}_{\text{OT}}$ (with $\varepsilon = 0$), is bounded by:*

$$W_1(f(\mathbf{x}), \mathbf{y}^*) \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}, \quad \forall \mathbf{p} \in \Pi.$$

Namely, $\mathbf{p}_{\text{OT}}$ minimizes the upper bound of $W_1(f(\mathbf{x}), \mathbf{y}^*)$, where the upper bound is based on the expected feature modification cost.

Although Theorem 4.1 bounds only the transport cost, that bound is a tractable convex proxy for the $\mathcal{NP}$ -hard discrete sparsity (L_0) problem. By concentrating attribution mass on the most efficient transport paths, the OT coupling steers the greedy selection in Algorithm 1 to discard non-essential features—the mechanism behind the large edit-count reductions observed empirically in Section 6. In addition, p -SHAP is conceptually correct in attributing causal behavior to the feature modifications, as stated in Theorem 4.2 below (proof in Appendix D).

Theorem 4.2 (Interventional Effect of p -SHAP). *For any subset $S \subseteq \mathcal{F}$ and any $\mathbf{x}_i$ ($i = 1, 2, \dots, n$), $v^{(i)}(S)$ represents the causal effect of the difference between the expected value of $f(\mathbf{r})$ under the intervention on features S and the unconditional expected value of $f(\mathbf{r})$. Mathematically, this is expressed as:*

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

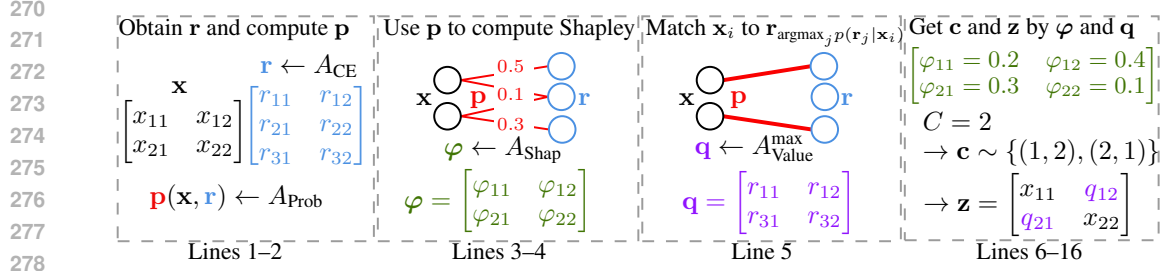


Figure 2: [An illustration of COLA] How COLA produces the action indicator $\mathbf{c}$ and the refined counterfactual $\mathbf{z}$ for equation (1). For clarity we use the simpler $A_{\text{Value}}^{\max}$ in line 5. In lines 6–16 we set the edit budget to $C = 2$; the sampling then selects exactly two positions to modify according to the attribution matrix $\boldsymbol{\varphi}$, illustrating how a tight budget concentrates edits on the highest-attribution positions.

Furthermore, p -SHAP preserves the desirable axioms of B-SHAP and RB-SHAP, so its empirical advantages come at no cost in attribution soundness; it retains every guarantee the baselines enjoy while adding the OT-coupling benefit. We omit the proof and refer to (Sundararajan & Najmi, 2020; Lundberg & Lee, 2017) for the Shapley axioms.

5 THE ALGORITHMIC FRAMEWORK COLA

Sketch The algorithmic framework COLA, stated in Algorithm 1 below, targets equation (1) and is built from four categories of interchangeable algorithms. This modularity is what makes the head-to-head comparison in Section 6 clean: swapping only the coupling A_{Prob} or the Shapley rule A_{Shap} isolates the source of each measured gain. We explain Algorithm 1 in details below, along with an illustration in Figure 2.

Algorithm 1 COUNTERFACTUALS WITH LIMITED ACTIONS (COLA)

Require: Model f , factual $\mathbf{x} \in \mathbb{R}^{n \times d}$, target $\mathbf{y}^* \in \mathbb{R}^m$, ϵ , and C
Ensure: Action plan $\mathbf{c} \in \mathbb{R}^{n \times d}$ and correspondingly a refined counterfactual $\mathbf{z} \in \mathbb{R}^{n \times d}$

- 1: Use $A_{\text{CE}}(f, \mathbf{x}, \mathbf{y}^*, \epsilon)$ to obtain $\mathbf{r} \in \mathbb{R}^{m \times d}$, with $\mathbf{y}^* = f(\mathbf{r})$ and $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$.
- 2: Use $A_{\text{Prob}}(\mathbf{x}, \mathbf{r})$ to obtain the joint distribution matrix $\mathbf{p} \in \mathbb{R}_+^{n \times m}$.
- 3: Use $A_{\text{Shap}}(\mathbf{x}, \mathbf{r}, \mathbf{p})$ to obtain the shapley values $\boldsymbol{\phi} \in \mathbb{R}^{n \times d}$ for each element of $\mathbf{x}$.
- 4: Normalize the element-wise absolute values of $\boldsymbol{\phi}$, i.e., $\varphi_{ik} \leftarrow |\phi_{ik}| / \|\boldsymbol{\phi}\|_1$ ($\boldsymbol{\varphi} \in \mathbb{R}_+^{n \times d}$).
- 5: Use $A_{\text{Value}}(\mathbf{r}, \mathbf{p})$ to obtain matrix $\mathbf{q} \in \mathbb{R}^{n \times d}$.
- 6: For $\mathbf{c} \in \{0, 1\}^{n \times d}$, $c_{ik} \leftarrow 0$ ($i = 1, \dots, n$, $k = 1, \dots, d$).
- 7: Sample C pairs (i, k) according to the probability matrix $\boldsymbol{\varphi}$, and let $c_{ik} = 1$ for them.
- 8: Let $\mathbf{z} \leftarrow \mathbf{x}$ ($\mathbf{z} \in \mathbb{R}^{n \times d}$).
- 9: **for** $i \leftarrow 1$ to n **do**
- 10: **for** $k \leftarrow 1$ to d **do**
- 11: **if** $c_{ik} = 1$ **then**
- 12: $z_{ik} \leftarrow q_{ik}$
- 13: **end if**
- 14: **end for**
- 15: **end for**
- 16: **return** $\mathbf{c}$ and $\mathbf{z}$

Line 1 (Applying a CE algorithm to find a counterfactual $\mathbf{r}$). The CE algorithm A_{CE} takes the model f , the factual $\mathbf{x}$, the target outcome $\mathbf{y}^*$, and the tolerance ϵ as input. The algorithm returns a counterfactual $\mathbf{r}$ staying close with $\mathbf{x}$, with $\mathbf{y}^* = f(\mathbf{r})$ and $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$.

Line 2 (Seeking a joint distribution of $\mathbf{x}$ and $\mathbf{r}$). We use an algorithm A_{Prob} for this task, which takes $\mathbf{x}$ and $\mathbf{r}$ as input, and outputs a matrix representing the joint distribution of all n and m data points in $\mathbf{x}$ and $\mathbf{r}$, respectively. The joint distribution $\mathbf{p}$ represents an alignment relationship (or

matching) between the factual rows and counterfactual rows, and we use it in Line 5 to compute the values that can be used for composing $\mathbf{z}$ later on. As discussed in Section 3, A_{Prob} can be based on OT to compute a joint distribution that yields the smallest OT distance between $\mathbf{x}$ and $\mathbf{r}$, if the alignment relationship between their rows are unknown. Otherwise, it is recommended to select a joint distribution that accurately reflects the alignment between the rows in $\mathbf{x}$ and $\mathbf{r}$.

Lines 3–4 (*p-SHAP FA*). We apply equation (7) to compute the Shapley values for $\mathbf{x}$. The joint distribution $\mathbf{p}$ can (optionally) align each row of $\mathbf{x}$ with its corresponding counterfactual rows of $\mathbf{r}$, so that the selected rows of $\mathbf{r}$ act as the most representative contrastive reference. This alignment is not a minor detail: numerically it is the single largest driver of contrastive FA quality, and the gap between the OT coupling and the uniform/random couplings in Figure 3 is attributable precisely to it. Finally, the Shapley-value matrix of $\mathbf{x}$ is taken element-wise in absolute value and normalized to sum to one, yielding the attribution matrix φ .

Line 5 (*Computing feature values*). The algorithm A_{Value} is used for this task, which takes the counterfactual $\mathbf{r}$ and the joint distribution $\mathbf{p}$ as input. For any row i in $\mathbf{x}$, A_{Value} selects one or multiple row(s) in $\mathbf{r}$, used as references for making changes in $\mathbf{x}$. The algorithm returns a matrix $\mathbf{q} \in \mathbb{R}^{n \times d}$, where each element q_{ik} serves as a counterfactual candidate for x_{ik} ($\forall i, k$). Below, we introduce $A_{\text{Value}}^{\text{max}}$ and $A_{\text{Value}}^{\text{avg}}$, respectively for the cases of selecting single row and selecting multiple rows.

For any row $\mathbf{x}_i$, $A_{\text{Value}}^{\text{max}}$ selects the row of $\mathbf{r}$ with the highest probability.

$$\mathbf{q} = A_{\text{Value}}^{\text{max}}(\mathbf{r}, \mathbf{p}), \text{ where } q_{ik} = r_{\tau(i), k} \text{ and } \tau(i) = \arg \max_{j=1, 2, \dots, m} p_{ij}. \quad (9)$$

The algorithm $A_{\text{Value}}^{\text{avg}}$ computes q_{ik} as a convex combination as a weighted average of $r_{1k}, r_{2k}, \dots, r_{mk}$.

$$\mathbf{q} = A_{\text{Value}}^{\text{avg}}(\mathbf{r}, \mathbf{p}), \text{ where } q_{ik} = \sum_{j=1}^m \left(\frac{p_{ij}}{\sum_{j'=1}^m p_{ij'}} \right) r_{jk}. \quad (10)$$

Lines 6–16. Recall that the non-negative matrix φ is normalized to have its summation being one. We could hence treat it as a policy to select the positions in $\mathbf{x}$ for value replacement (i.e. $c_{ik} = 1$), as what line 7 does. Then, for any i and k with $c_{ik} = 1$, x_{ik} gets modified to q_{ik} , and the modified matrix is then returned as $\mathbf{z}$ together with $\mathbf{c}$ forming the optimized solutions of the problem in equation (1).

By Theorem 4.1, COLA is designed to shrink the dissimilarity between $f(\mathbf{z})$ and $\mathbf{y}^*$ by editing $\mathbf{z}$ only at the feature positions that attribution flags as most important for the counterfactual effect. The theorem below (proof in Appendix E) establishes a proximity guarantee: the refined $\mathbf{z}$ produced by COLA satisfies the constraint equation (1b) in the typical scenario where $n = m$ under the Frobenius norm, never drifting farther from the factual input than the aligned proposal, so refinement can only tighten proximity. The empirical counterpart is the relative-displacement column of Table 3, whose values—as low as 13.4%—show the refined counterfactuals sitting far closer to the factuals than the originals across all datasets.

Theorem 5.1 (Counterfactual Proximity). *Let $\mathbf{q} \in \mathbb{R}^{n \times d}$ be the aligned reference matrix derived from the counterfactual set $\mathbf{r} \in \mathbb{R}^{m \times d}$ via the value assignment step $\mathbf{q} = A_{\text{Value}}(\mathbf{r}, \mathbf{p})$, where $\mathbf{p}$ is the coupling matrix. For any dimensions $n, m \geq 1$, the refined counterfactual $\mathbf{z}$ constructed by the COLA framework satisfies:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

This inequality guarantees that the refined counterfactual $\mathbf{z}$ is strictly closer (or equal) to the factual input $\mathbf{x}$ compared to the aligned counterfactual proposal $\mathbf{q}$, ensuring that the refinement process introduces no additional divergence.

Corollary 5.2. *In the special case where $n = m$ and the OT plan $\mathbf{p}$ corresponds to a deterministic permutation σ (i.e., obtained without entropic regularization, $\varepsilon = 0$), the matrix $\mathbf{q}$ becomes a row-permuted version of $\mathbf{r}$ (denoted as $\mathbf{r}_\sigma$, where $q_i = r_{\sigma(i)}$). In this setting, the bound simplifies to:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{r}_\sigma - \mathbf{x}\|_F,$$

recovering the intuition that the refined counterfactual is at least as close to the factuals as the original counterfactual set reordered by σ .

Complexity of COLA The favorable edit-count numbers do not come at a steep computational price. Let $\mathcal{O}(M_{\text{CE}})$ be the complexity of A_{CE} , and for A_{Shap} let M_{Shap} be the number of sampled subsets in a weighted-linear-regression Shapley estimator. The complexity of COLA in n, m, d , and the entropic-OT regularizer ε is $\mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + N$ where $N = \mathcal{O}(nm) + \mathcal{O}(nd)$ if $A_{\text{Value}}^{\max}$ is used and $N = \mathcal{O}(nmd)$ if $A_{\text{Value}}^{\text{avg}}$ is used. See Appendix F for more details.

6 NUMERICAL RESULTS

This section evaluates COLA on the problem in equation (1), with $\mathbf{y}^* = f(\mathbf{r})$ where $\mathbf{r}$ is the counterfactual from a CE method A_{CE} . To ensure the comparison is not an artifact of any single distance, we report results under four divergences: OT (distance between full distributions), MMD (mean discrepancy in a high-dimensional feature space), and the absolute mean difference (MeanD) and absolute median difference (MedianD) (mean and median differences). Every claim below is measured against these baselines under identical target effects, so that only the number of edits varies. The three findings, stated as testable quantitative claims, are: I) *COLA achieves the target effect with far fewer edits than the original CE (a $2 \times -4 \times$ reduction)*. II) *p-SHAP beats all five competing Shapley couplings, and does so in every scenario*. III) *COLA comes within a small margin of the exact MILP optimum where that optimum is computable*.

Experiment Setup Our evaluation grid is deliberately large so that the reported margins reflect breadth rather than a favorable pick: 4 binary-classification datasets, 5 CE algorithms spanning diverse goals, and 12 classifiers (Table 1). Each (dataset, A_{CE} , model) triple defines one “experiment scenario,” giving hundreds of distinct scenarios in which the six methods of Table 2 are compared.

Table 1: [Experiment Scenarios Setup] The comparison grid: four datasets $\times$ five CE algorithms $\times$ twelve models f . A “scenario” is one (dataset, A_{CE} , model) combination; every method in Table 2 is scored on the same scenarios so results are directly comparable.

Dataset	HELOC (FICO, 2018), German Credit (Hofmann, 1994), Hotel Bookings (Antonio et al., 2019), COMPAS (Jeff Larson et al., 2016)
A_{CE}	DiCE (Mothilal et al., 2020), AReS (Rawal & Lakkaraju, 2020), GlobeCE (Ley et al., 2023), KNN (Albini et al., 2022; Contardo et al.; Forel et al., 2023), Discount (You et al., 2025)
Model f	Bagging, LightGBM, Support Vector Machine (SVM), Gaussian Process (GP), Radial Basis Function Network (RBF), XGBoost, Deep Neural Network (DNN), Random Forest (RndForest), AdaBoost, Gradient Boosting (GradBoost), Logistic Regression (LR), Quadratic Discriminant Analysis (QDA)

Table 2: [Experiment Methods Setup] The 6 compared methods, color-coded to match Figure 3. The proposed p -SHAP is CF- p_{OT} (red); the remaining five are baselines. Methods differ only in the coupling $\mathbf{p}$ and the Shapley rule A_{Shap} , so any performance gap is attributable to those two choices alone. Each method is run in the shared scenarios of Table 1.

Method	The probability $\mathbf{p}$ used by A_{Value} and A_{Shap}	The Shapley algorithm A_{Shap}
RB- p_{Uni}	$\mathbf{p} \leftarrow A_{\text{Prob}}: p_{ij} = 1/nm \ (\forall i, j)$	RB-SHAP, $\mathcal{D} = \text{trainset of } f$
RB- p_{OT}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Eq. equation (8) (but not used in } A_{\text{Shap}})$	RB-SHAP, $\mathcal{D} = \text{trainset of } f$
CF- p_{Uni}	$\mathbf{p} \leftarrow A_{\text{Prob}}: p_{ij} = 1/nm \ (\forall i, j)$	CF-SHAP, $\mathcal{D}(\mathbf{x}_i) = \mathbf{p}_i$
CF- p_{Rnd}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Any } \mathbf{x}_i \text{ matched randomly to an } \mathbf{r}_j$	CF-SHAP, $\mathcal{D}(\mathbf{x}_i) = \mathbf{p}_i$
CF- p_{OT}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Eq. equation (8)}$	p -SHAP with $\mathbf{p}$
CF- p_{Ect}	$\mathbf{p} \leftarrow A_{\text{Prob}}: \text{Any } \mathbf{x}_i \text{ matched to known counterpart } \mathbf{r}_j$	(CF or B)-SHAP, $\mathcal{D}(\mathbf{x}_i) \rightarrow \mathbf{r}_j$

The five A_{CE} in Table 1 deliberately span the regimes discussed earlier: DiCE and KNN are instance-based (one or more counterfactuals per factual instance); AReS and GlobeCE are group-based (a shared counterfactual set for the whole factual group); and Discount treats the factuals as an empirical distribution and seeks a nearby counterfactual distribution. Covering all three regimes is what lets us claim the gains are generator-agnostic rather than tuned to one CE style.

Table 2 defines the 6 compared methods—CF- p_{OT} is the proposed p -SHAP, the other five are baselines—each determined by a choice of A_{Prob} , A_{Value} , and A_{Shap} . For instance, RB- p_{Uni} uses a uniform A_{Prob} for $\mathbf{p}$, feeds that $\mathbf{p}$ into A_{Value} via equation (9) or equation (10) to form $\mathbf{q}$, and sets A_{Shap} to RB-SHAP.

The six methods form a controlled ablation ladder in which each rung isolates one variable. (i) RB- p_{Uni} vs. CF- p_{Uni} isolates the value of using CE information at all. (ii) CF- p_{Uni} , CF- p_{Rnd} , and CF- p_{OT} share the CE information but differ in the coupling, isolating the contribution of the OT alignment—which, we show, dominates the uniform and random couplings by a wide margin. (iii) CF- p_{OT} vs. RB- p_{OT} differs only in A_{Shap} , confirming that OT helps *because* it informs the attribution with a factual-counterfactual alignment, not through some incidental effect. (iv) CF- p_{Ect} is the privileged reference in which each counterfactual has a *known* exact source, giving an exact alignment (so CF-SHAP coincides with B-SHAP)—a strong yardstick that, remarkably, CF- p_{OT} still matches or beats in Result III.

Table 3: [COLA for Modification Minimality] Number of modified features in $\mathbf{z}$ (lower is better) needed to reach 80% and 100% of the counterfactual effect of $f(\mathbf{r})$, together with the relative displacement $\|\mathbf{z} - \mathbf{x}\|/\|\mathbf{r} - \mathbf{x}\|$ (lower is better). Each entry averages 4 randomly selected scenarios from Table 1 with 100 runs each ($\pm$ std. shown); “-” marks methods that cannot reach the target effect at all. **Bold** marks the best method per block: the proposed CF- p_{OT} (p -SHAP) is best in every row of every dataset, and is the only method able to attain the 100% effect anywhere in the table.

Dataset	Method	80% Counterfactual Effect		100% Counterfactual Effect	
		# Modified Features	$\frac{\ \mathbf{z} - \mathbf{x}\ }{\ \mathbf{r} - \mathbf{x}\ }$	# Modified Features	$\frac{\ \mathbf{z} - \mathbf{x}\ }{\ \mathbf{r} - \mathbf{x}\ }$
German Credit $ \mathcal{F} = 9$	RB- p_{Uni}	—	—	—	—
	RB- p_{OT}	5.29(± 0.09)	75.9%	—	—
	CF- p_{Uni}	—	—	—	—
	CF- p_{Rnd}	—	—	—	—
	CF- p_{OT}	1.70(± 0.02)	24.3%	3.13(± 0.03)	44.9%
Hotel Bookings $ \mathcal{F} = 29$	RB- p_{Uni}	7.10(± 0.08)	24.5%	—	—
	RB- p_{OT}	8.55(± 0.08)	50.2%	—	—
	CF- p_{Uni}	7.01(± 0.07)	41.1%	—	—
	CF- p_{Rnd}	10.63(± 0.08)	62.4%	—	—
	CF- p_{OT}	2.50(± 0.03)	14.6%	4.44(± 0.02)	26.0%
COMPAS $ \mathcal{F} = 15$	RB- p_{Uni}	5.02(± 0.05)	82.7%	—	—
	RB- p_{OT}	—	—	—	—
	CF- p_{Uni}	2.80(± 0.04)	34.4%	—	—
	CF- p_{Rnd}	2.58(± 0.04)	32.1%	—	—
	CF- p_{OT}	1.25(± 0.03)	14.8%	2.45(± 0.03)	30.0%
HELOC $ \mathcal{F} = 23$	RB- p_{Uni}	—	—	—	—
	RB- p_{OT}	—	—	—	—
	CF- p_{Uni}	—	—	—	—
	CF- p_{Rnd}	2.73(± 0.04)	15.7%	—	—
	CF- p_{OT}	2.35(± 0.03)	13.4%	7.745(± 0.05)	44.7%

Result I: COLA reaches the target effect with a $2 \times -4 \times$ cut in edits In Table 3 we require $\mathbf{z}$ to reach 80% or 100% of the counterfactual effect of $f(\mathbf{r})$ ² and read off the edit count of the refined $\mathbf{z}$ at that matched effect. The reductions are large and hold across all four datasets: relative to the original CE $\mathbf{r}$, the refined $\mathbf{z}$ needs far fewer edits. At the 80% level, the p -SHAP counterfactual $\mathbf{z}$ moves only 13.4%–24.3% of the original distance $\|\mathbf{r} - \mathbf{x}\|$ (column $\|\mathbf{z} - \mathbf{x}\|/\|\mathbf{r} - \mathbf{x}\|$)—i.e. COLA discards roughly three-quarters or more of the original displacement while retaining 80% of the effect. The contrast at the 100% level is sharper still: CF- p_{OT} (p -SHAP) is the *only* method that attains the full effect on any dataset, and it does so using just 26.0%–44.9% of the feature changes in the original $\mathbf{r}$ —a roughly $2 \times$ to $4 \times$ reduction. Every competing method leaves the 100% column empty. The following analysis isolates *why* p -SHAP achieves this best-in-class edit minimality.

²By definition $D(f(\mathbf{r}), \mathbf{y}^*) = 0$ is the 100% effect (since $\mathbf{y}^* = f(\mathbf{r})$). The 80% level is the fraction of divergence removed by the refined CE $\mathbf{z}$, i.e. $Counterfactual\ Effect = 1 - D(f(\mathbf{z}), \mathbf{y}^*)/D(f(\mathbf{x}), \mathbf{y}^*) = 80\%$, with D being OT.

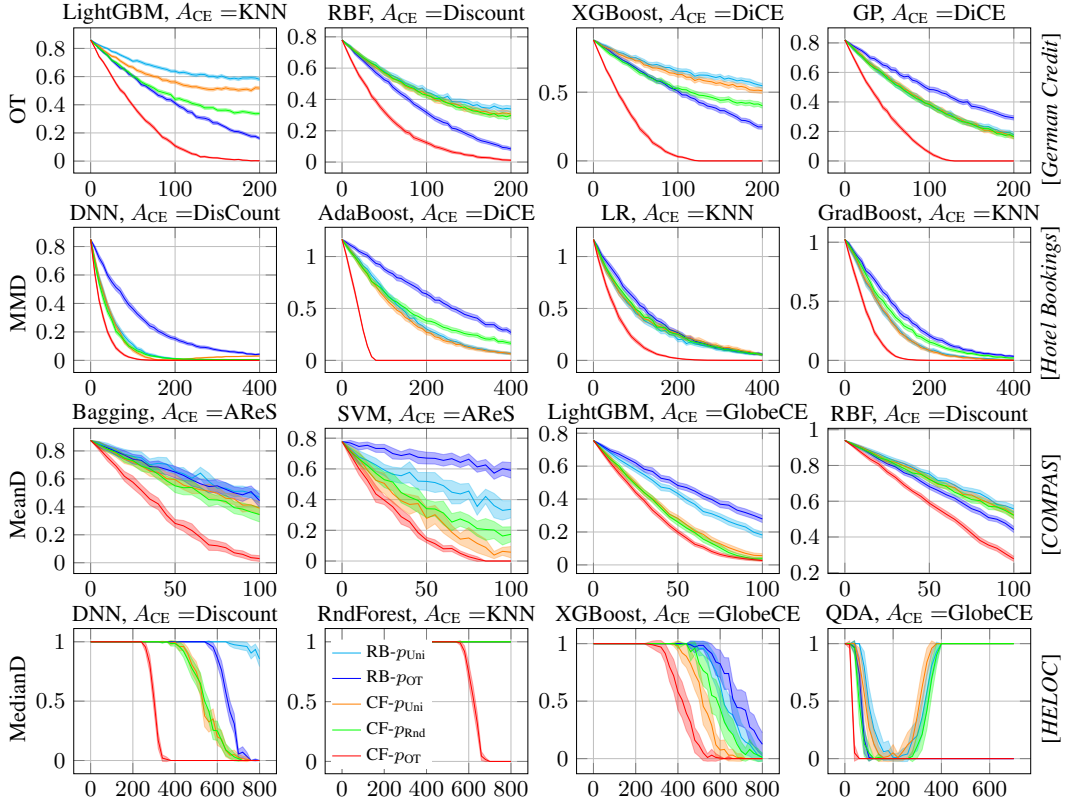


Figure 3: Residual effect gap $D(f(z), y^*)$ (lower is better) vs. the edit budget C ; a curve that drops faster and lower is the better method. Across all panels the red CF- p_{OT} (p -SHAP) curve lies at or below every competitor at essentially every budget, and the CE-agnostic RB- p_{Uni} /RB- p_{OT} curves are the highest. Each point averages 100 runs; shaded bands are 99.9% confidence intervals (narrow bands indicate the ranking is stable, not noise). A_{Value}^{avg} is used for *HELOC* and *COMPAS*, A_{Value}^{max} for *German Credit* and *Hotel Bookings*. The legend in “RnDForest, $A_{CE} = KNN$ ” applies to all plots.

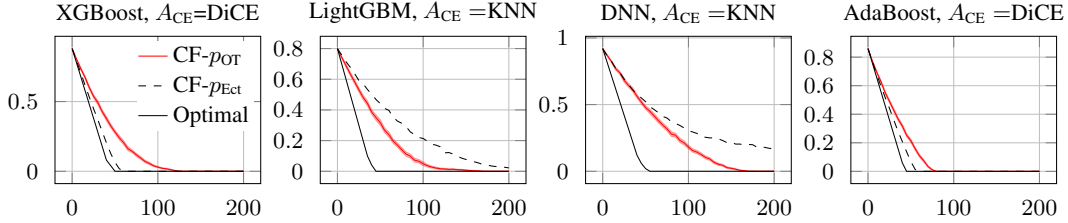


Figure 4: [*German Credit*] Residual effect gap $D(f(z), y^*)$ vs. edit budget C under $D = MeanD$, with the exact MILP optimum overlaid as the lower reference. CF- p_{OT} and the known-alignment CF- p_{Ect} both hug the optimal curve, and CF- p_{OT} at times matches or dips below CF- p_{Ect} despite using no privileged alignment—quantifying how small the remaining gap to optimality is.

Result II: p -SHAP outperforms every competing Shapley coupling, and the ablation pinpoints why Figure 3 plots the residual effect gap $D(f(z), y^*)$ of equation (1) against the edit budget C for the Shapley methods of equation (4)–equation (7). The controlled ablation of Table 2 lets us read each margin as a single-variable effect. First, the two CE-agnostic couplings RB- p_{Uni} and RB- p_{OT} trail all others by a wide, consistent margin—using CE information is a prerequisite, not a refinement. Second, comparing CF- p_{OT} with RB- p_{OT} —which differ *only* in A_{Shap} —the former is uniformly lower, isolating the OT alignment (rather than any incidental factor) as the source of the gain. Third, CF- p_{OT} decisively beats CF- p_{Uni} and CF- p_{Rnd} , the two variants that also use CE information but with uniform or random couplings; this shows that possessing counterfactual information is insufficient and the

quality of the alignment is what matters—and, as Figure 4 later shows, the best alignment need not be the exact CE-generation one.

The dominance is not scenario-specific: CF- p_{OT} is the top method across the scenarios of Table 1 in both Figure 3 and Table 3, and a much larger battery of scenarios in Appendix H reproduces the same ordering. In sum, p -SHAP delivers the sparsest CE of every Shapley method compared, in every scenario tested.

Result III: COLA lands close to the exact MILP optimum—and the alignment-free p -SHAP even rivals the known-alignment reference This experiment quantifies how much p -SHAP loses by discarding CE-generation knowledge³ and substituting the OT coupling for the CE-algorithm-dependent $\mathcal{D}$, as shown in Figure 4. We run CF- p_{Ect} inside COLA on equation (1) with $D = \text{MeanD}$; because CF- p_{Ect} uses a known alignment, it is the natural upper reference for what alignment information can buy. The exact optimum here is obtained by solving an MILP (derivation in Appendix G); solving it is computationally heavy, so we report it for *German Credit*.

The gap to optimality is small: CF- p_{Ect} tracks the exact optimum closely under DiCE (near-optimal). Since the DiCE and KNN pairs are independent across instances, this also shows COLA is effective for instance-based CE, even though equation (1) was formulated for the more general group/distributional case. The most striking comparison is that CF- p_{OT} *sometimes matches or surpasses* CF- p_{Ect} despite using no privileged alignment, because its OT-based selection of the features to modify rests on a firmer theoretical footing than the CE-algorithm-dependent alignment, whose value for FA is not guaranteed. A residual gap between both methods and the optimum remains under KNN, so identifying the truly best alignment is still open—and quantifies the headroom left for future work.

7 CONCLUSIONS

This paper introduced COLA, a framework that refines CE through joint-distribution-informed Shapley values (p -SHAP) so the refined CE preserves the counterfactual effect while requiring far fewer edits. The empirical case is broad and consistent: across four datasets, twelve models, and five CE generators, p -SHAP reproduces the target effect using only 26%–45% of the original feature changes—a 2× to 4× reduction—is the best of six methods in every scenario of Table 3, and is the only method to reach the full 100% effect on all datasets. On a tractable benchmark, COLA further sits within a small margin of the exact MILP optimum, with the alignment-free p -SHAP occasionally matching the known-alignment reference. Together these margins position p -SHAP as a practical, generator-agnostic default for sparse counterfactual explanations.

ETHICS STATEMENT

All authors have read and agree to abide by the ICLR Code of Ethics.⁴ This paper proposes a method (*COLA* with p -SHAP) for refining CE. Our work is methodological and empirical; it does not involve human subjects or interventions, and we did not collect, annotate, or release any new personal data.

All experiments use standard, de-identified, publicly available tabular datasets (e.g., Adult/Census Income, German Credit, HELOC). We follow dataset licenses/terms and the common research practice of using these data solely for benchmarking. We do not attempt re-identification or linkage, and we release no additional personal information. Our code will include options to exclude protected attributes and to enforce feasibility/immunity constraints on features.

CE can surface biases in underlying models but do not by themselves guarantee fairness. Our method can reduce the number of recommended feature edits; it does *not* authorize edits to sensitive or immutable attributes (e.g., sex, race, age) nor does it confer causal validity. We discourage use of CE to “game” high-stakes systems (e.g., lending, hiring, healthcare) or to pressure individuals into unrealistic or unethical behavior changes. Any deployment in consequential domains should include domain-expert-defined feasibility constraints, fairness audits (e.g., disparate impact/benefit across

³This knowledge—the exact factual-counterfactual alignment—is available only for DiCE and KNN among the CE methods considered.

⁴<https://iclr.cc/public/CodeOfEthics>

groups), human oversight, and compliance with applicable data-protection and AI regulations. We explicitly caution that our attribution is distributional (not causal) and must not be interpreted as causal effect.

Risks include (i) misinterpretation of attributions as causal, (ii) recommending infeasible or harmful edits, and (iii) enabling strategic manipulation. We mitigate these by (a) stating non-causal scope and limitations in the paper, (b) supporting cost/feasibility masks and immutable features in the implementation, (c) reporting performance across subgroups where relevant, and (d) providing guidance for responsible use in documentation.

We provide an anonymized implementation, detailed settings, and seeds to facilitate replication; we report model types, hyperparameters, and compute budgets, and we avoid result cherry-picking by using fixed evaluation protocols. To our knowledge, there are no conflicts of interest or sensitive sponsorships related to this submission; any funding disclosures will be provided after the double-blind review.

REPRODUCIBILITY STATEMENT

We take reproducibility seriously and provide pointers to all necessary components. The COLA framework and p-SHAP attribution are specified with pseudocode and notation in the main paper (Algorithm 1; Method and Preliminaries sections), with all theoretical assumptions and complete proofs deferred to the appendix (theorems and lemmas with line-by-line proofs).

Implementation details—hyperparameters, optimization settings, environment specifications, and random seeds—are documented in Appendix I, which also describes the datasets and the evaluation protocol (data splits and preprocessing). The six compared methods, their coupling choices, and every reported number rest on the shared scenarios and the fixed 100-run, 99.9%-confidence-interval protocol described in the experimental sections and ablations, with configurations mirrored in the supplemental materials so each margin can be reproduced. The experiment code and software will be released upon acceptance; we report hardware and runtime in Appendix I to contextualize compute, and provide ablation/sensitivity studies and the small-scale MILP-optimum benchmark to support independent verification of the reported gains.

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A COMPARISON TO EXISTING APPROACHES

CF-SHAP (Albini et al., 2022) uses counterfactual data points as the Shapley background distribution, but assumes that distribution is defined conditionally on each single instance—i.e. a *known* probabilistic alignment between every factual and every counterfactual. Similar assumptions appear in (Merrick & Taly, 2020; Kommiya Mothilal et al., 2021), and they break down in the many global-explanation settings; this is exactly the assumption our OT coupling removes, and our experiments quantify the benefit of doing so (CF- p_{OT} vs. the CE-conditioned CF- p_{Uni} /CF- p_{Rnd}). The setup of (Albini et al., 2022) is a special case of ours. WeightedSHAP (Kwon & Zou, 2022) weights features rather than treating them equally; our method instead weights data points and extends naturally to weighting both rows and columns. The framework of (Kommiya Mothilal et al., 2021) combines FA and CE, but inherits the alignment assumption of (Albini et al., 2022) and thus does not generalize to group (Ley et al., 2023; 2022) or distributional CE (You et al., 2025). The causal-structure/time-series decomposition of (Sharma et al.) attributes a single observed system-level change to individual inputs. Crucially, none of these works targets the minimal-actions CE problem or reports edit-count reductions at matched effect—the quantity on which our method is evaluated.

The problems in (Kanamori et al., 2022; Karimi et al., 2021) are closest to ours. Karimi et al. (2021) minimize action cost under a known structural causal model (SCM), which is rarely available in practice. Kanamori et al. (2022) note it remains open whether existing CE methods can solve that problem, and propose an MILP-solver approach for linear classifiers, tree ensembles, and deep ReLU networks, building on (Ustun et al., 2019; Kanamori et al., 2020; Parmentier & Vidal, 2021). Because solving MILP is costly and scales poorly, we use it not as a competitor but as an exact-optimum yardstick on a small benchmark (Result III), against which COLA’s near-optimal, far cheaper solutions are measured.

B $\mathcal{NP}$ -HARDNESS OF THE PROBLEM IN EQUATION (1)

Theorem B.1 below states that, unless $\mathcal{P} = \mathcal{NP}$, no scalable exact algorithm is expected in general; the hardness stems not only from the non-linearity of f but also from the combinatorial edit-selection structure. This is what makes COLA’s near-optimal performance in Result III noteworthy: it approaches an exact optimum that is provably intractable to compute at scale.

Theorem B.1. *Problem equation (1) is generally $\mathcal{NP}$ -hard for non-trivial divergences. More specifically, it is hard even when $d = 1$ for linear models.*

Proof. Consider the *Sparse Regression (SR) with a Cardinality Constraint* problem, defined as follows. Given a matrix $\mathbf{W} \in \mathbb{R}^{m \times n}$, a target vector $\mathbf{y}^* \in \mathbb{R}^m$, and a sparsity level $K \in \mathbb{N}$, the goal is to find a vector $\mathbf{z} \in \mathbb{R}^n$ that minimizes the residual error while having at most K non-zero

elements:

$$\begin{aligned} \min_{\mathbf{z}} \quad & \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2, \\ \text{s.t.} \quad & \|\mathbf{z}\|_0 \leq K, \end{aligned}$$

where $\|\mathbf{z}\|_0$ denotes the number of non-zero elements in $\mathbf{z}$. This problem is known to be $\mathcal{NP}$ -hard due to the combinatorial nature of selecting the subset of variables to include in the model.

We will map this SR problem to our problem equation (1) with the following settings. Let the number of features be $d = 1$ and the number of instances be n (the same as the dimension of the SR problem). Let the factual data be $\mathbf{x} = \mathbf{0} \in \mathbb{R}^n$ (the zero vector), and the target model output be $\mathbf{y}^* \in \mathbb{R}^m$ (as given in the SR problem). We define the model f as a linear function $f(\mathbf{z}) = \mathbf{W}\mathbf{z}$.

For the model output, we define $D(f(\mathbf{z}), \mathbf{y}^*) = \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2$, i.e. the Euclidean distance squared. For the instances, we define $D(\mathbf{z}, \mathbf{x}) = \|\mathbf{z} - \mathbf{x}\|_2^2 = \|\mathbf{z}\|_2^2$, since $\mathbf{x} = \mathbf{0}$. We set the maximum allowed feature changes C to be equal to k (the sparsity level from the SR problem). The large constant M can be any sufficiently large positive number, for example, $M \geq \max_i |z_i|$.

Given that $\mathbf{x} = \mathbf{0}$, constraints equation (1d) and equation (1e) are simplified to:

$$\begin{aligned} z_i &\leq 0 \cdot (1 - c_i) + M c_i = M c_i, \\ z_i &\geq 0 \cdot (1 - c_i) - M c_i = -M c_i, \quad \forall k = 1, \dots, n. \end{aligned}$$

This means that if $c_i = 0$, then $z_i \leq 0$ and $z_i \geq 0$, so $z_i = 0$. If $c_i = 1$, then $z_i \in [-M, M]$. The constraint equation (1c) becomes $\sum_{k=1}^n c_i \leq k$.

Our problem equation (1) thus becomes:

$$\min_{\mathbf{c}, \mathbf{z}} \quad \|\mathbf{W}\mathbf{z} - \mathbf{y}^*\|_2^2, \tag{11a}$$

$$\text{s.t.} \quad \|\mathbf{z}\|_2^2 \leq \epsilon, \tag{11b}$$

$$\sum_{k=1}^d c_i \leq k, \tag{11c}$$

$$z_i = 0, \quad \text{if } c_i = 0, \tag{11d}$$

$$z_i \in [-M, M], \quad \text{if } c_i = 1, \tag{11e}$$

$$c_i \in \{0, 1\}, \quad \forall k = 1, \dots, n. \tag{11f}$$

In this formulation, the variables c_i indicate whether the variable z_i is allowed to change ($c_i = 1$) or not ($c_i = 0$). The constraints enforce that $z_i = 0$ when $c_i = 0$, mirroring the sparsity constraint in the SR problem. The constraint $\sum_{k=1}^n c_i \leq k$ ensures that at most k features can change, matching the sparsity level. The objective function is identical to that of SR.

Since our problem formulation directly mirrors the SR problem with a cardinality constraint, which is known to be $\mathcal{NP}$ -hard, solving Problem equation (1) is at least as hard as solving the SR problem. Therefore, Problem equation (1) is $\mathcal{NP}$ -hard even when $d = 1$, the model f is linear, and the divergence functions D are standard Euclidean distances. $\square$

C PROOF OF THEOREM 4.1: p -SHAP TOWARDS COUNTERFACTUAL EFFECT

Let $\mathbf{x} = \{\mathbf{x}_i\}_{i=1}^n \in \mathbb{R}^{n \times d}$ be the set of factual data points with associated probability weights $\mu_i \geq 0$ such that $\sum_{i=1}^n \mu_i = 1$, and let $\mathbf{r} = \{\mathbf{r}_j\}_{j=1}^m \in \mathbb{R}^{m \times d}$ be the set of counterfactual data points with associated probability weights $\nu_j \geq 0$ such that $\sum_{j=1}^m \nu_j = 1$. Let $\mathbf{p}_{\text{OT}} \in \mathbb{R}^{n \times m}$ be the OT plan between $\mathbf{x}$ and $\mathbf{r}$ that minimizes the expected transportation cost:

$$\mathbf{p}_{\text{OT}} = \arg \min_{\mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu})} \sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2,$$

where $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$ is the set of joint distributions satisfying the marginal constraints $\sum_{j=1}^m p_{ij} = \mu_i$ for all i and $\sum_{i=1}^n p_{ij} = \nu_j$ for all j . The theorem below shows that the feature attributions align with the expected feature-modification costs, so the resulting action plans are cost-efficient in reaching the counterfactual outcome—the theoretical basis for the large, consistent edit-count reductions measured in Section 6.

Theorem C.1 (Theorem 4.1 in the main text). *Consider the 1-Wasserstein divergence W_1 , i.e. $W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*|$. Suppose the counterfactual outcome $\mathbf{y}^*$ is fully achieved by $\mathbf{r}$, i.e. $\mathbf{y}_j^* = f(\mathbf{r}_j)$ ($j = 1, 2, \dots, m$). Assume that the model $f : \mathbb{R}^d \rightarrow \mathbb{R}$ is Lipschitz continuous with Lipschitz constant L . The expected absolute difference in model outputs between the factual and counterfactual instances, weighted by $\mathbf{p}_{OT}$ (with $\varepsilon = 0$), is bounded by:*

$$W_1(f(\mathbf{x}), \mathbf{y}^*) \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \quad \forall \mathbf{p} \in \Pi.$$

Namely, $\mathbf{p}_{OT}$ minimizes the upper bound of $W_1(f(\mathbf{x}), \mathbf{y}^*)$, where the upper bound is based on the expected feature modification cost.

Proof. Since the model f is Lipschitz continuous with constant L , for any $\mathbf{x}_i \in \mathbb{R}^d$ and $\mathbf{r}_j \in \mathbb{R}^d$, it holds that:

$$|f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Multiplying both sides of the inequality by $p_{ij}^{OT} \geq 0$, we obtain:

$$p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Summing both sides over all $i = 1, \dots, n$ and $j = 1, \dots, m$, we have:

$$\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)| \leq L \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2.$$

Let us denote $E_f = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} |f(\mathbf{x}_i) - f(\mathbf{r}_j)|$ and $E_d = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2$. The inequality then becomes:

$$E_f \leq L E_d.$$

To further bound E_d , we apply the Cauchy-Schwarz inequality. Observe that the weights p_{ij}^{OT} are non-negative and satisfy $\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} = 1$ because $\mathbf{p}_{OT}$ is a probability distribution over the joint space of $\mathbf{x}$ and $\mathbf{r}$. The Cauchy-Schwarz inequality states that for any real-valued functions a_{ij} and b_{ij} ,

$$\left(\sum_{i,j} a_{ij} b_{ij} \right)^2 \leq \left(\sum_{i,j} a_{ij}^2 \right) \left(\sum_{i,j} b_{ij}^2 \right).$$

Setting $a_{ij} = \sqrt{p_{ij}^{OT}}$ and $b_{ij} = \sqrt{p_{ij}^{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2$, we have:

$$E_d = \sum_{i,j} p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2 = \sum_{i,j} \sqrt{p_{ij}^{OT}} \sqrt{p_{ij}^{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2 = \sum_{i,j} a_{ij} b_{ij}.$$

Applying the Cauchy-Schwarz inequality:

$$E_d^2 \leq \left(\sum_{i,j} a_{ij}^2 \right) \left(\sum_{i,j} b_{ij}^2 \right) = \left(\sum_{i,j} p_{ij}^{OT} \right) \left(\sum_{i,j} p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 \right).$$

Since $\sum_{i,j} p_{ij}^{OT} = 1$, this simplifies to:

$$E_d^2 \leq \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2 = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2.$$

Taking the square root of both sides, we obtain:

$$E_d \leq \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{OT} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Substituting back into the inequality for E_f , we have:

$$E_f \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Note that the 1-Wasserstein divergence is no more than E_f ⁵, then,

$$W_1(f(\mathbf{x}), \mathbf{y}^*) = \min_{\pi \in \Pi} \sum_{i=1}^n \sum_{j=1}^m \pi_{ij} |f(\mathbf{x}_i) - \mathbf{y}_j^*| \leq E_f \leq L \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}.$$

Therefore, the 1-Wasserstein divergence between $f(\mathbf{x})$ and $\mathbf{y}^*$ is bounded by the Lipschitz constant L times the square root of the expected transportation cost under $\mathbf{p}_{\text{OT}}$. Since $\mathbf{p}_{\text{OT}}$ minimizes the expected transportation cost $\sum_{i,j} p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2$ over all feasible transport plans in $\Pi(\boldsymbol{\mu}, \boldsymbol{\nu})$, we have

$$\sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2} \leq \sqrt{\sum_{i=1}^n \sum_{j=1}^m p_{ij} \|\mathbf{x}_i - \mathbf{r}_j\|_2^2}, \quad \forall \mathbf{p} \in \Pi(\boldsymbol{\mu}, \boldsymbol{\nu}).$$

Hence the conclusion follows. $\square$

D PROOF OF THEOREM 4.2: INTERVENTIONAL EFFECT

Theorem D.1 (Theorem 4.2 in the main text). *For any subset $S \subseteq \mathcal{F}$ and any $\mathbf{x}_i$ ($i = 1, 2, \dots, n$), $v^{(i)}(S)$ represents the causal effect of the difference between the expected value of $f(\mathbf{r})$ under the intervention on features S and the unconditional expected value of $f(\mathbf{r})$. Mathematically, this is expressed as:*

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Proof. Let $p(\mathbf{x}, \mathbf{r})$ be the joint probability of $\mathbf{x}$ and $\mathbf{r}$ obtained from A_{Prob} . Under the intervention $do(\mathbf{r}_S = \mathbf{x}_{i,S})$, the features in S are set to $\mathbf{x}_{i,S}$, and the features in $\mathcal{F} \setminus S$ remain distributed according to their marginal distribution $p(\mathbf{r}_{\mathcal{F} \setminus S})$. Therefore, the expected value of $f(\mathbf{r})$ under the intervention is:

$$\mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})] = \int_{\mathcal{R}_{\mathcal{F} \setminus S}} f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S}) p(\mathbf{r}_{\mathcal{F} \setminus S}) d\mathbf{r}_{\mathcal{F} \setminus S}.$$

Remark that by the definition of $v^{(i)}(S)$,

$$v^{(i)}(S) = \mathbb{E}_{\mathbf{r} \sim p(\mathbf{r} | \mathbf{x}_i)}[f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S})] - \mathbb{E}[f(\mathbf{r})] = \int_{\mathcal{R}_{\mathcal{F} \setminus S}} f(\mathbf{x}_{i,S}; \mathbf{r}_{\mathcal{F} \setminus S}) p(\mathbf{r}_{\mathcal{F} \setminus S}) d\mathbf{r}_{\mathcal{F} \setminus S} - \mathbb{E}[f(\mathbf{r})],$$

such that

$$\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S) = \mathbb{E}[f(\mathbf{r}) | do(\mathbf{r}_S = \mathbf{x}_{i,S})].$$

Hence the conclusion. $\square$

The value function $v^{(i)}(S)$ captures the expected value of $f(\mathbf{r})$ when we intervene by setting the features in S to $\mathbf{x}_{i,S}$, denoted as $do(\mathbf{r}_S = \mathbf{x}_{i,S})$. This intervention is independent of any predefined joint probability distribution $p(\mathbf{x}, \mathbf{r})$. Therefore, the expression $\mathbb{E}[f(\mathbf{r})] + v^{(i)}(S)$ represents the combined effect of the base expected value of $f(\mathbf{r})$ and the additional causal impact of the attribution $v^{(i)}(S)$.

⁵Note that $E_f = \sum_{i=1}^n \sum_{j=1}^m p_{ij}^{\text{OT}} |f(\mathbf{x}_i) - f(\mathbf{r}_j)|$. Because both π and $\mathbf{p}_{\text{OT}}$ denote joint probability of $\mathbf{x}$ and $\mathbf{r}$, however, π makes the summation the minimum for the W_1 term across all possible joint distributions, whereas $\mathbf{p}_{\text{OT}}$ does not.

E PROOF OF THEOREM 5.1: COUNTERFACTUAL PROXIMITY

Theorem E.1 (Theorem 5.1 in the main text). *Let $\mathbf{q} \in \mathbb{R}^{n \times d}$ be the aligned reference matrix derived from the counterfactual set $\mathbf{r} \in \mathbb{R}^{m \times d}$ via the value assignment step $\mathbf{q} = A_{\text{value}}(\mathbf{r}, \mathbf{p})$, where $\mathbf{p}$ is the coupling matrix. For any dimensions $n, m \geq 1$, the refined counterfactual $\mathbf{z}$ constructed by the COLA framework satisfies:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

This inequality guarantees that the refined counterfactual $\mathbf{z}$ is strictly closer (or equal) to the factual input $\mathbf{x}$ compared to the aligned counterfactual proposal $\mathbf{q}$, ensuring that the refinement process introduces no additional divergence.

Proof. Let $\mathbf{q}$ be the aligned reference matrix generated by $A_{\text{value}}(\mathbf{r}, \mathbf{p})$. For the elements where $c_{ik} = 1$ (modified elements), $z_{ik} = q_{ik}$. For elements where $c_{ik} = 0$, $z_{ik} = x_{ik}$. Therefore, we can write:

$$(z_{ik} - x_{ik})^2 = \begin{cases} (q_{ik} - x_{ik})^2, & \text{if } c_{ik} = 1; \\ 0, & \text{if } c_{ik} = 0. \end{cases} \quad (12)$$

The squared Frobenius norms with respect to $\mathbf{q}$ and $\mathbf{z}$ are computed as follows:

$$\begin{aligned} \|\mathbf{q} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2, \\ \|\mathbf{z} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (z_{ik} - x_{ik})^2. \end{aligned}$$

And,

$$\begin{aligned} \|\mathbf{q} - \mathbf{x}\|_F^2 - \|\mathbf{z} - \mathbf{x}\|_F^2 &= \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2 - \sum_{i=1}^n \sum_{k=1}^d (z_{ik} - x_{ik})^2 \\ &\stackrel{(i)}{=} \sum_{i=1}^n \sum_{k=1}^d (q_{ik} - x_{ik})^2 - \sum_{(i,k): c_{ik}=1} (q_{ik} - x_{ik})^2 \\ &= \sum_{(i,k): c_{ik}=0} (q_{ik} - x_{ik})^2 \geq 0, \end{aligned}$$

where the equality (i) holds because of equation (12).

Since the difference $\|\mathbf{q} - \mathbf{x}\|_F^2 - \|\mathbf{z} - \mathbf{x}\|_F^2 \geq 0$, it follows that:

$$\|\mathbf{z} - \mathbf{x}\|_F^2 \leq \|\mathbf{q} - \mathbf{x}\|_F^2.$$

Taking square roots, we get:

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F.$$

Hence the conclusion. $\square$

Corollary E.2 (Corollary 5.2 in the main text). *In the special case where $n = m$ and the OT plan $\mathbf{p}$ corresponds to a deterministic permutation σ (i.e., obtained without entropic regularization, $\varepsilon = 0$), the matrix $\mathbf{q}$ becomes a row-permuted version of $\mathbf{r}$ (denoted as $\mathbf{r}_\sigma$, where $q_i = r_{\sigma(i)}$). In this setting, the bound simplifies to:*

$$\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{r}_\sigma - \mathbf{x}\|_F,$$

recovering the intuition that the refined counterfactual is at least as close to the factuals as the original counterfactual set reordered by σ .

Proof. Under the assumptions $n = m$ and $\varepsilon = 0$, the optimal transport plan $\mathbf{p}$ becomes a permutation matrix associated with a bijection σ . Consequently, the value assignment algorithm A_{value} (whether maximizing or averaging) assigns $\mathbf{q}_i = \mathbf{r}_{\sigma(i)}$ for every instance i . Substituting $\mathbf{q}$ with the permuted matrix $\mathbf{r}_\sigma$ into the general bound established in Theorem 5.1 (i.e., $\|\mathbf{z} - \mathbf{x}\|_F \leq \|\mathbf{q} - \mathbf{x}\|_F$) directly yields the result. $\square$

F ANALYSIS OF COMPUTATIONAL COMPLEXITY OF COLA

We perform analysis of the computational complexity of COLA as follows.

First, we analyze A_{Prob} . If the alignment between $\mathbf{x}$ and $\mathbf{r}$ is known a priori, then A_{Prob} just constructs the matrix $\mathbf{p}$ with the prior knowledge, which takes $\mathcal{O}(n \times m)$. Otherwise, we consider solving OT to obtain $\mathbf{p}$, which, by the Sinkhorn–Knopp algorithm, takes $\mathcal{O}(n \times m \times \log(1/\varepsilon))$.

Then we analyze A_{Shap} . For each subset, the model is evaluated on all n data points, leading to $\mathcal{O}(n)$ evaluations per subset. Incorporating baseline values from the reference data $\mathbf{r}$ involves replacing the values of certain features with their corresponding baseline values. This operation is $\mathcal{O}(m)$ because it requires accessing the baseline values from the reference table $\mathbf{r}$ for each of the d features. If we assume that the reference values can be precomputed and accessed in constant time, then the complexity of incorporating these values can be considered as $\mathcal{O}(d)$. The number of M_{Shap} subsets results in M_{Shap} model evaluations. Combining the above steps, the complexity of A_{Shap} is $\mathcal{O}(n \times d \times M_{\text{Shap}})$.

The normalization in line 4 of COLA takes $\mathcal{O}(n \times d)$.

To compare the complexities of the two algorithms, $A_{\text{Value}}^{\text{max}}$ and $A_{\text{Value}}^{\text{avg}}$, we analyze each algorithm step-by-step. For $A_{\text{Value}}^{\text{max}}$, for each row $\mathbf{x}_i$ in the data table, we need to (1) compute the probabilities p_{ij} for all $j \in \{1, 2, \dots, m\}$, which involves $\mathcal{O}(m)$ operations per row, (2) identify the row $\mathbf{r}_j$ in the reference data with the highest probability p_{ij} , which involves $\mathcal{O}(m)$ operations per row, and (3) assign $q_{ik} = r_{\tau(i),k}$ where $\tau(i) = \arg \max_j p_{ij}$, which involves $\mathcal{O}(d)$ operations per row. Since there are n rows in the data table, the total complexity for $A_{\text{Value}}^{\text{max}}$ is $\mathcal{O}(n \times (m + m + d)) = \mathcal{O}(n \times (2m + d)) = \mathcal{O}(n \times m + n \times d) = \mathcal{O}(n \times (m + d))$.

For $A_{\text{Value}}^{\text{avg}}$, for each row $\mathbf{x}_i$ in the data table, we need to 1) compute the probabilities p_{ij} for all $j \in \{1, 2, \dots, m\}$, which involves $\mathcal{O}(m)$ operations per row, 2) compute the sum $\sum_{j'=1}^m p_{ij'}$, which involves $\mathcal{O}(m)$ operations per row, and 3) calculate the weighted average $q_{ik} = \sum_{j=1}^m \left(\frac{p_{ij}}{\sum_{j'=1}^m p_{ij'}} \right) r_{jk}$, which involves $\mathcal{O}(m \times d)$ operations per row. Since there are n rows in the data table, the total complexity for $A_{\text{Value}}^{\text{avg}}$ is $\mathcal{O}(n \times (m + m + m \times d)) = \mathcal{O}(n \times (2m + m \times d)) = \mathcal{O}(n \times (m + m \times d)) = \mathcal{O}(n \times m \times (1 + d)) = \mathcal{O}(n \times m \times d)$.

For lines 6–16, the entire complexity is straightforwardly $\mathcal{O}(n \times d) + \mathcal{O}(C) = \mathcal{O}(n \times d)$ due to the fact $C \leq n \times d$.

Therefore, the complexity of COLA using $A_{\text{Value}}^{\text{max}}$ equals

$$\begin{aligned} & \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nd) + \mathcal{O}(n(m + d)) + \mathcal{O}(nd) \\ &= \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nm) + \mathcal{O}(nd), \end{aligned}$$

and the complexity of COLA using $A_{\text{Value}}^{\text{avg}}$ equals

$$\begin{aligned} & \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nd) + \mathcal{O}(nmd) + \mathcal{O}(nd) \\ &= \mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + \mathcal{O}(nmd). \end{aligned}$$

Hence the complexity of COLA with respect to n , m , d , and the regularization parameter ε of entropic OT is

$$\mathcal{O}(M_{\text{CE}}) + \mathcal{O}(nm \log(1/\varepsilon)) + \mathcal{O}(ndM_{\text{Shap}}) + N,$$

where $N = \mathcal{O}(nm) + \mathcal{O}(nd)$ if $A_{\text{Value}}^{\text{max}}$ is used and $N = \mathcal{O}(nmd)$ if $A_{\text{Value}}^{\text{avg}}$ is used.

G AN MILP FORMULATION OF EQUATION (1) WITH MEAND

This section derives the exact global-optimum benchmark against which COLA is compared in Result III, for the case of a known factual-counterfactual alignment solving equation (1) with D being MeanD, namely

$$D(f(\mathbf{z}), \mathbf{y}^*) = \left| \frac{1}{n} \sum_{i=1}^n f(\mathbf{z}_i) - \bar{y}^* \right|,$$

with $\bar{y}^* = \frac{1}{m} \sum_{j=1}^m y_j^*$. Since COLA is used, we have $D(\mathbf{r}, \mathbf{x}) \leq \epsilon$, and $\mathbf{z}$ stays closer to $\mathbf{x}$ than $\mathbf{r}$, hence equation (1b) is dropped. The formulation of equation (1) then becomes:

$$\min_{\mathbf{c}, \mathbf{z}} \left| \sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^* \right|, \quad (13a)$$

$$\text{s.t.} \quad \sum_{i=1}^n \sum_{k=1}^d c_{ik} \leq C, \quad (13b)$$

$$z_{ik} = r_{ik}c_{ik} + x_{ik}(1 - c_{ik}), \quad i = 1, \dots, n, \quad k = 1, \dots, d. \quad (13c)$$

Note that the original constraints equation (1d) and equation (1e) merge to be equation (13c), because CF- p_{Ect} is imposed to be used. That is, for any $\mathbf{x}_i$, there is an exact $\mathbf{r}_j$ serves as its reference in A_{Shap} and A_{Value} , such that x_{ik} ($k = 1, 2, \dots, d$) either stays unchanged or can be changed to r_{jk} . Therefore $z_{ik} = r_{ik}c_{ik} + x_{ik}(1 - c_{ik})$ of which the value depends on the binary variable c_{ik} .

Due to the known alignment between any $\mathbf{x}_i$ and its corresponding $\mathbf{r}_j$, $\mathbf{q}$ is determined (also, both $A_{\text{Values}}^{\max}$ and $A_{\text{Values}}^{\text{avg}}$ return the same $\mathbf{q}$). For any data point i and any feature set $S \subseteq \mathcal{F}$, let $\mathbf{z}_{iS}$ denotes the solution $\mathbf{z}_i$ where we have all features $k \in S$ changed to q_{ik} , and the other features $h \in \mathcal{F} \setminus S$ stays x_{ih} . Hence the set of $\mathbf{z}_{iS}$ ($S \subseteq \mathcal{F}$) composes the domain of all possible values of $\mathbf{z}$. Define a corresponding scaler variable for any $\mathbf{z}_{iS}$:

$$g_{iS} = f(\mathbf{z}_{iS}) - \bar{y}^*.$$

Then, for any $\mathbf{z}_i$ in equation (13), the value of the term $\sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^*$ can be represented by a binary variable a_{iS} together with the scaler g_{iS} , namely,

$$\sum_{i=1}^n f(\mathbf{z}_i) - n\bar{y}^* = \sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS}.$$

The optimization problem equation (13) is hence reformulated as a mixed integer programming below.

$$\min_{\mathbf{a}, \eta} \quad \eta, \quad (14a)$$

$$\text{s.t.} \quad \sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS} \leq \eta, \quad (14b)$$

$$\sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} g_{iS} a_{iS} \geq -\eta, \quad (14c)$$

$$\sum_{S \subseteq \mathcal{F}} a_{iS} = 1, \quad i = 1, \dots, n, \quad (14d)$$

$$\sum_{i=1}^n \sum_{S \subseteq \mathcal{F}} |S| a_{iS} \leq C. \quad (14e)$$

Minimizing η under the two constraints equation (14b) and equation (14c) is equivalent to minimizing the objective function of equation (13). The constraints in equation (14d) guarantees that each data point i is subject to one and only one feature modification plan a_{iS} for a specific S ($S \subseteq \mathcal{F}$). The constraint equation (14e) corresponds to equation (13b). Solving equation (14) yields the exact optimum for COLA under a known factual-counterfactual alignment—the lower-reference curve in Figure 4 of Section 6, which lets us report how small COLA’s gap to optimality is.

H EXTENDED NUMERICAL RESULTS

The extended grid in Figures 5–8 reproduces the main-paper ordering across many more scenarios. Read as pairwise ablations, two comparisons stand out. Holding everything but A_{Shap} fixed, CF- p_{Uni} beats RB- p_{Uni} ; likewise CF- p_{OT} consistently beats RB- p_{OT} . Both comparisons point the same way and quantify a single conclusion: *RB-SHAP is not suitable for FA in CE.*

We analyze how the Shapley choice (lines 3–4 of COLA) affects FA through two single-variable comparisons. First, $RB-p_{Uni}$ vs. $CF-p_{Uni}$ differ only in A_{Shap} ; across Figures 5–8, $CF-p_{Uni}$ is generally the lower (better) curve. Second, $RB-p_{OT}$ vs. $CF-p_{OT}$ also differ only in A_{Shap} , and the latter is consistently below the former. The two pairwise gaps together isolate the CE-aware Shapley rule as the driver, confirming that *RB-SHAP is not suitable for FA in CE*.

Besides FA, the other equally important step of COLA is line 5: using the joint probability $p(\mathbf{x}, \mathbf{r})$ to build $\mathbf{q}$, which tells the factual x in which direction to move its features toward the target outcome. Here the evidence is unambiguous— $CF-p_{OT}$ is the lowest curve in *all* experiments of Figures 5–8, without exception. The three couplings $CF-p_{Uni}$, $CF-p_{Rnd}$, and $CF-p_{OT}$ all supply a joint probability $\mathbf{p}$ when the exact alignment is unknown, yet their curves separate sharply. The quantitative takeaway is that simply knowing the CE (and its marginal distribution) is not enough; the OT coupling adds a large, consistent margin on top.

OT is most valuable precisely where alignment information is missing or unreliable. Strikingly, even when the CE algorithm explicitly pairs each factual with a counterfactual, that known alignment is not demonstrably optimal for performance—Figure 4 in Section 6 shows $CF-p_{OT}$ occasionally matching or beating the known-alignment $CF-p_{Ect}$.

This clarifies that p_{OT} need not be the true data-generating joint distribution of $\mathbf{x}$ and $\mathbf{r}$; its role is to guide COLA to treat $\mathbf{x}$ and $\mathbf{r}$ jointly for both FA and CE. As a robustness signal, the QDA column of Figure 5 shows the OT-based methods staying stable and on-target while the others diverge markedly from the target. We stress that COLA is *not limited to OT* as A_{Prob} : Figure 4 indicates that even the best known $\mathbf{p}$ leaves a non-negligible gap to the global optimum, so this remaining margin marks a concrete opportunity for a stronger A_{Prob} in future work.

I EXPERIMENTS REPRODUCIBILITY

The experiments run on a high-performance computing (HPC) cluster of four nodes (one per dataset) in parallel, each node with two Intel Xeon 2660v3 CPUs (10 cores, 2.60GHz) and 128 GB memory. Each node takes roughly 5–10 hours depending on dataset size; the experiments also reproduce on a laptop, at higher wall-clock cost. These modest requirements underscore that the reported edit-count gains are obtained without heavy compute.

For the four datasets, numerical features are standardized and categorical features use label- or one-hot encoding. The two encodings yielded no remarkable difference in COLA’s performance, so the reported margins are not an artifact of encoding choice. The train-test split is 7 : 3.

The exact optimum in Figure 4 is solved with Gurobi 11.0.2 ([gur](#)) (a license is required; the open-source Google OR-Tools ([goo](#)) is an alternative). Because the MILP of Appendix G is computationally expensive, this exact reference is practical only for small datasets such as German Credit—which is precisely why it serves as a controlled optimality check rather than a scalable method, and why COLA’s ability to approach it cheaply is the salient comparison. Computing the optimum on larger datasets would require reducing the number of features used.

To keep the comparison fair and free of per-model tuning, model hyperparameters are left at their defaults wherever possible. Bagging, GP, RBF, RndForest, AdaBoost, GradBoost, and QDA use the scikit-learn implementations ([skl](#)) with default hyperparameters. DNN, SVM, RBF, and LR are implemented in PyTorch ([Paszke et al., 2019](#)); the DNN has three layers and the SVM uses a linear kernel. XGBoost ([Chen & Guestrin, 2016](#)) and LightGBM ([Ke et al., 2017](#)) are used through their scikit-learn interface, again with defaults. Since none of the twelve models is tuned to favor COLA, the uniform edit-count advantages reported across them reflect the method rather than model selection.

THE USE OF LARGE LANGUAGE MODELS (LLMs)

We used large language models (LLMs) only as general-purpose assist tools. Their role was limited to grammar, wording, and light copy-editing of author-written text. LLMs did not contribute research ideation, modeling choices, experimental design, or results. All algorithms, proofs, datasets, and analyses were created and verified by the authors. Any LLM-suggested phrasing was reviewed and

1188 edited before inclusion in the paper. The authors take full responsibility for all content; LLMs are not
1189 authors or contributors. This disclosure complies with the ICLR policy on LLM usage.
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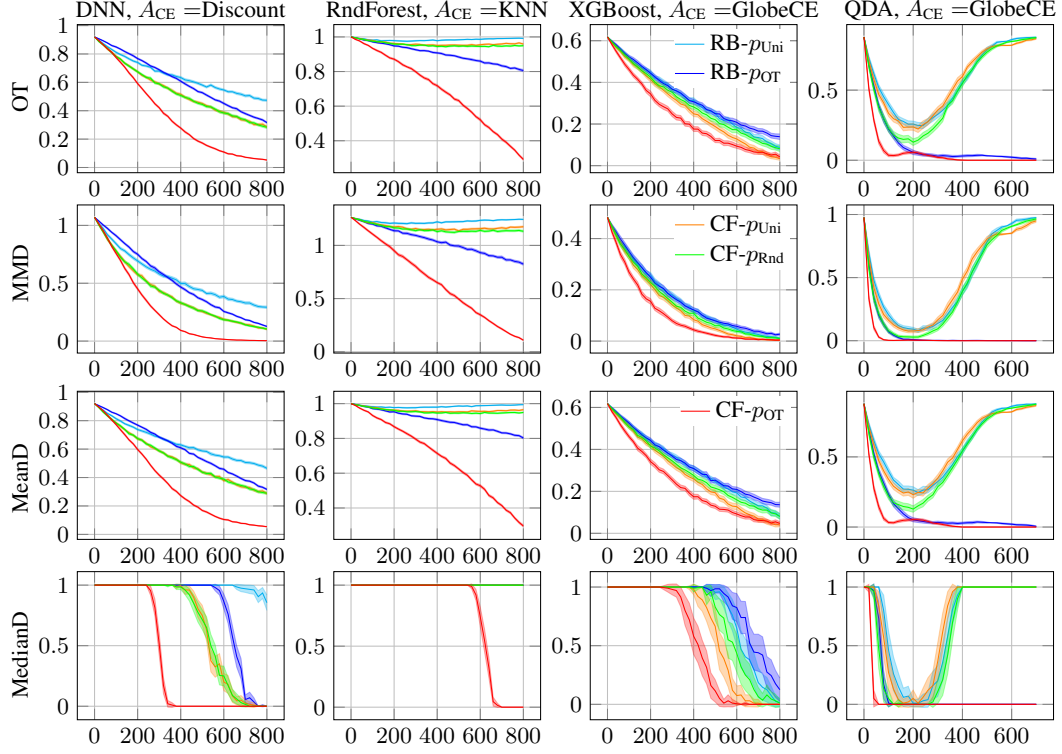


Figure 5: [HELOC] Residual effect gap $D(f(\mathbf{z}), \mathbf{y}^*)$ (lower is better) vs. edit budget C over many (model, A_{CE} , divergence) scenarios; CF- p_{OT} (p -SHAP) is the lowest curve throughout. Each point averages 100 runs; shaded bands are 99.9% confidence intervals. Legends apply to all plots. A_{Value}^{avg} is used.

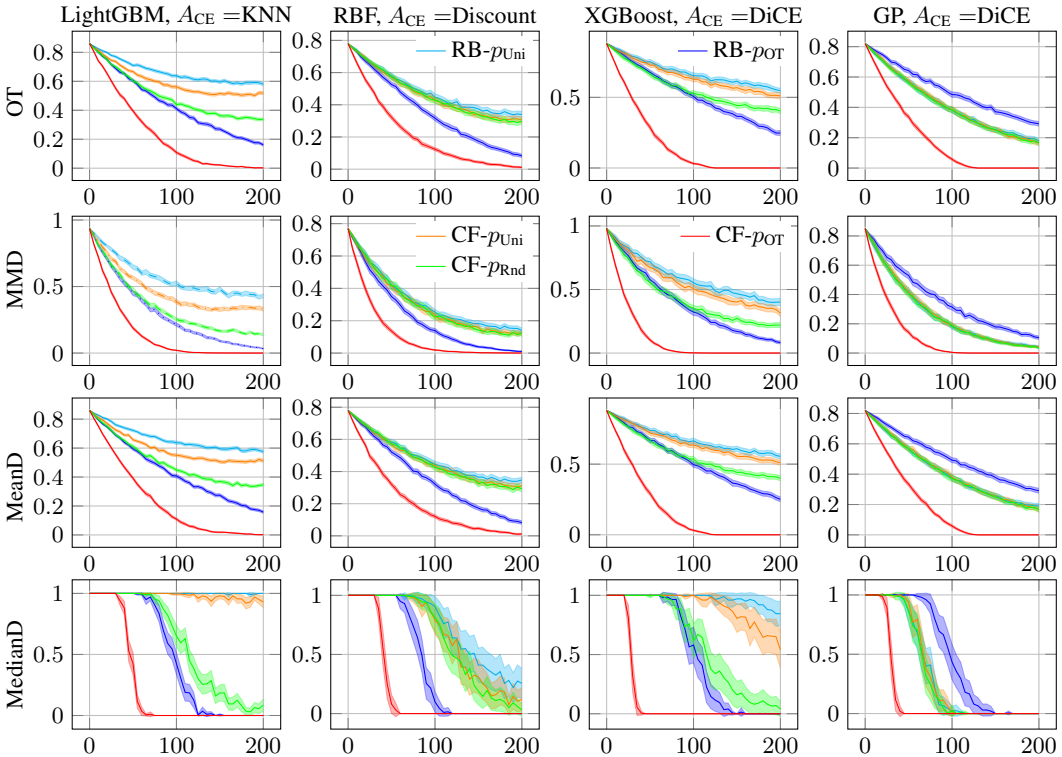


Figure 6: [German Credit] Residual effect gap $D(f(\mathbf{z}), \mathbf{y}^*)$ (lower is better) vs. edit budget C across scenarios; CF- p_{OT} (p -SHAP) again tracks lowest while the CE-agnostic RB couplings trail. Each point averages 100 runs; shaded bands are 99.9% confidence intervals. Legends apply to all plots. A_{Value}^{avg} is used.

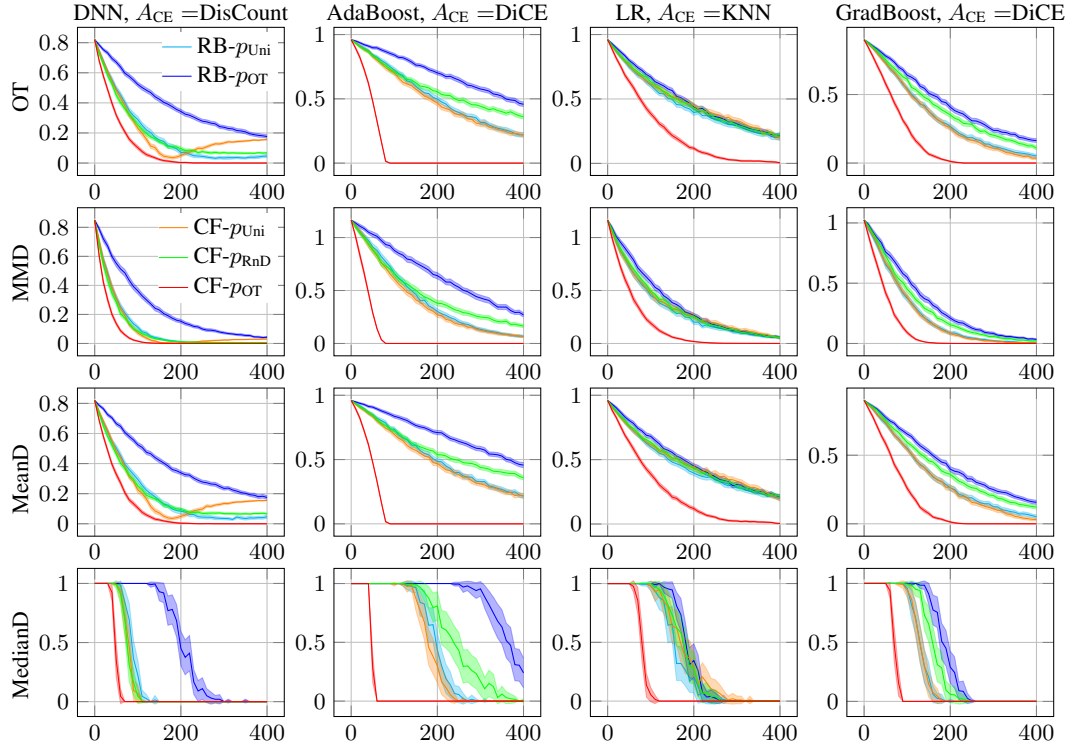


Figure 7: [Hotel Bookings] Residual effect gap $D(f(\mathbf{z}), \mathbf{y}^*)$ (lower is better) vs. edit budget C across scenarios; the CF- p_{OT} (p -SHAP) curve stays lowest, consistent with the largest edit-count reduction reported for this dataset. Each point averages 100 runs; shaded bands are 99.9% confidence intervals. Legends apply to all plots. $A_{Value}^{\max}$ is used.

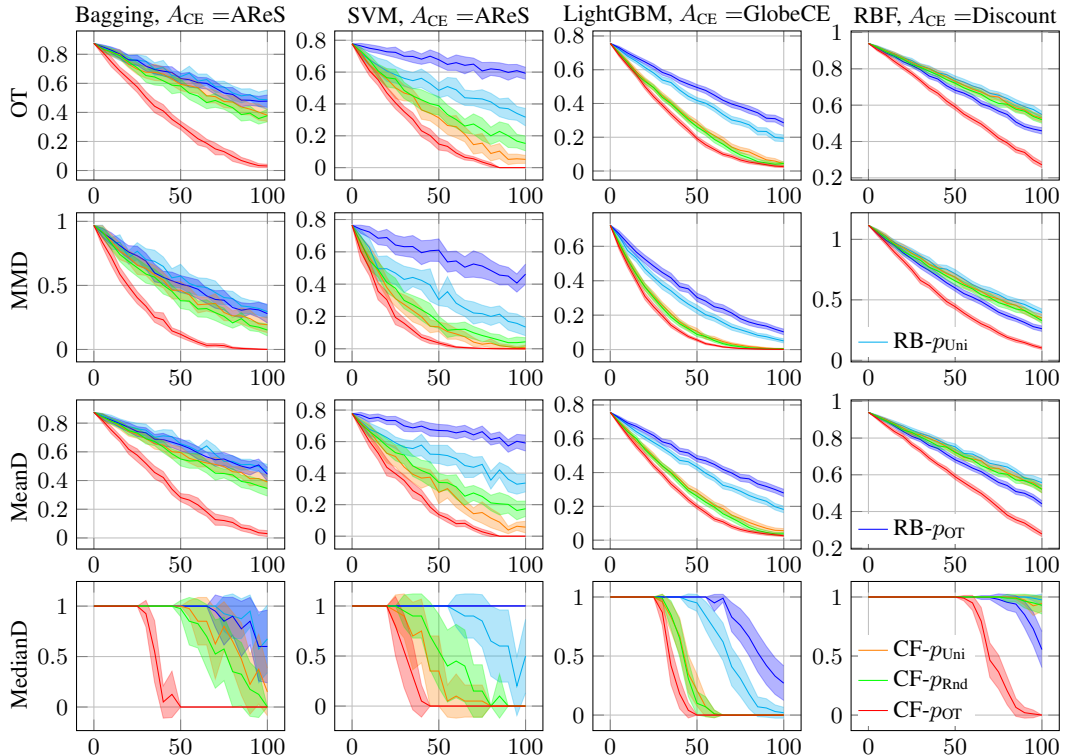


Figure 8: [COMPAS] Residual effect gap $D(f(\mathbf{z}), \mathbf{y}^*)$ (lower is better) vs. edit budget C across scenarios; CF- p_{OT} (p -SHAP) is the lowest curve throughout. Each point averages 100 runs; shaded bands are 99.9% confidence intervals. Legends apply to all plots. $A_{Value}^{\max}$ is used.